STRUCTURAL AND FUNCTIONAL CHARACTERIZATION OF THE MYCOBACTERIAL LIPOARABINOMANNAN SUCCINYLTRANSFERASE

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ABSTRACT

Mycobacterium tuberculosis (Mtb) is a medically important pathogen and is considered a major threat worldwide. The rise of drug-resistant Mtb has intensified the need for alternative therapeutic strategies and targeting bacterial virulence factors has emerged as a promising approach to combat antimicrobial resistance, driving renewed interest in the clinical potential of anti-virulence therapies. Mycobacteria possess an unusual, lipid and glycolipid-rich cell envelope that plays a central role in its pathogenicity, intrinsic drug resistance, and ability to evade host immune responses. This envelope is highly dynamic and can covalently modify its glycolipids to adapt to environmental stresses[1]. Lipoarabinomannan (LAM), a major glycoconjugate of the mycobacterial cell envelope, is a key virulence factor and immunomodulator essential for *M. tuberculosis* pathogenesis[2]. **Covalent modification of LAM by succinylation is known to regulate the mannose-capping, an essential process that pathogens use to evade the immune response of the host. Moreover, SucT is the sole enzyme that catalyses this reaction. If this protein is absent, the progressive branching of LAM is disrupted and alterations of cell envelope biogenesis occur**[2,3,4]. SucT is a key enzyme involved in the transmembrane succinylation of LAM. The molecular mechanisms underlying the translocation of succinyl groups remain largely unknown. Our team aims to decipher these mechanisms through a complementary multidisciplinary approach combining microbiology, enzymology, biophysics and structural biology cryo-EM, to obtain a detailed snapshot of the enzyme in action.

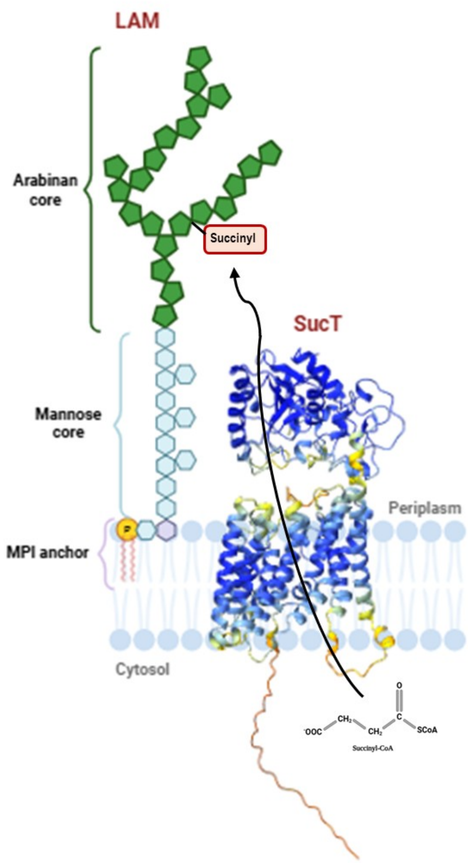

Fig. *Schematic representation of LAM glycan and Alphafold predicted structure of SucT.*

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